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An implementation of the AC-PGA metaheuristic for Depot-Free Multiple Traveling Salesperson Problems

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Summary

Depot-Free Multiple Traveling Salesperson Problems (DFmTSPs) are a family of $\mathcal{NP}$-hard optimization problems that generalize the classical Traveling Salesperson Problem (TSP) and the Multiple Traveling Salesperson Problem (mTSP). In a nutshell, a DFmTSP receives as input a complete edge-weighted graph $G = (V, E, w)$ and a positive integer m, with the purpose being to find a set of paths for m salespersons that visit all the vertices in $V(G)$ exactly once by minimizing an objective function (Cornejo-Acosta et al., 2023).

The Multiple Traveling Salesperson Problem and its extensions have many applications in logistics, transportation, scheduling, and other fields (Bektas, 2006; Cheikhrouhou & Khoufi, 2021). In the literature, multiple mTSP variants have been studied according to their characteristics. One of the features that has been used to categorize mTSP variants is the existence of depots. In this context, the depots are specially defined vertices from which salespersons depart. On the one hand, in the literature, mTSPs traditionally consider depots, and many studies in the literature focus on them. On the other hand, mTSPs that do not consider depots are known as DFmTSPs, and they have been studied less thoroughly (Cornejo-Acosta et al., 2023). The nature of the salespersons' paths is another feature used to categorize mTSP variants. The two main approaches are: closed paths, where the salespersons depart, visit a set of vertices, and return to the initial vertex by forming a cycle; and open paths, where the salespersons do not return to their initial vertices. Regarding the objective functions for mTSPs and DFmTSPs, two objective functions have been traditionally adopted: the `minsum` objective function, which minimizes the sum of the costs of the salespersons' paths, and the `minmax` objective function, which minimizes the longest path among the salespersons (Carter & Ragsdale, 2006). Among all the DFmTSP variants, the one that considers only the `minsum` objective function and closed paths is also known in the literature as the Hamiltonian p-median problem (HpMP) (Branco & Coelho, 1990). Although the HpMP and the mTSPs have some differences (mainly the concept of depot), some authors in the literature refer to them interchangeably.

In the literature, the closed-path DFmTSP with a `minsum` objective function (or simply HpMP) has been approached through integer programs (Karabulut et al., 2021), integer programs based on TSPs (Cornejo-Acosta et al., 2023), and other exact approaches like branch-and-cut algorithms (Barbato & Gouveia, 2024; Bektaş et al., 2019; Erdoğan et al., 2016). Besides, because of the complexity of the problem, many researchers have proposed alternatives that can execute in practical runtimes, such as heuristics (Cornejo-Acosta et al., 2025; Erdoğan et al., 2016), metaheuristics that are mostly based on advanced local search procedures, evolutionary

computing, or hybrid approaches (Erdoğan et al., 2016; Herrán et al., 2019; Jiang et al., 2020; Karabulut et al., 2021; Zhou et al., 2018).

This paper presents a heuristic solver for depot-free multiple traveling salesperson problems whose core engine is a metaheuristic algorithm from the literature known as `AC-PGA`. The `AC-PGA` (Ant Colony-Partheno Genetic Algorithm) is a metaheuristic algorithm that was initially proposed for the closed-path depot-free multiple traveling salesperson problem for the `minsum` objective function (Jiang et al., 2020). According to the nomenclature proposed by Cornejo-Acosta et al. (2023), this specific variant should be referred to as CP-DFmTSP, but it is also known just as the HpMP (Branco & Coelho, 1990). The `AC-PGA` is a hybrid metaheuristic that combines features of a Partheno Genetic Algorithm (`PGA`) and an Ant Colony-based optimization algorithm (`ACO`). Among the valuable features of the `AC-PGA` algorithm is that a new solution encoding method is proposed. According to the authors, this new solution encoding method is simpler and more efficient than those existing in the literature, and it also fits better with the nature of the problem. Another interesting feature is that the `AC-PGA` is capable of handling load-balance between the salespersons by limiting the minimum and maximum number of vertices for each salesperson's path. These latter constraints are known in the literature as bounding constraints (Kara & Bektas, 2006), and many proposals from the literature do not consider them. The authors carried out computational experimentation with benchmark instances from the TSPLIB library (Reinelt, 1991) with up to 280 vertices and up to 10 salespersons. The reported results show that the `AC-PGA` outperformed existing algorithms in the literature of the problem. Thus, this makes the `AC-PGA` a good reference to benchmark new proposals related to routing optimization problems. For a full description of the `AC-PGA`, the reader can refer to the original paper (Jiang et al., 2020).

Features of this AC-PGA software implementation

As mentioned in the previous section, the original `AC-PGA` was proposed for the HpMP, which is a particular case of the DFmTSP where the objective function is `minsum` and the paths are closed. Thus, since the scope is limited, its applicability may also be limited. In this software, we implemented additional features that extend the applicability of the `AC-PGA` to other variants of DFmTSPs.

- **Objective functions**: The software allows choosing not only the `minsum` objective function, but also the `minmax`. This may be of interest to researchers who want to validate new proposals by testing both objective functions. Furthermore, researchers working on similar optimization problems and applications aiming to minimize the "makespan" may also be interested.
- **Open paths**: The software allows choosing the kind of path (closed or open) to be used. In the literature, the open-path variants have not received as much attention as the closed-path variants, which may be indicative of a gap. This feature may be of interest to practitioners dealing with use cases involving open paths, or researchers interested in benchmarking new proposals against a validated and tested state-of-the-art metaheuristic.
- **Diversity track**: The implementation allows tracking the diversity of the population through the evolution of the generations. In the evolutionary computing literature, diversity refers to the number of different solutions present in a population. Thus, this feature may be of interest to researchers who intend to delve deeper into the behavior of diversity in evolutionary computing to design and propose techniques that can improve the performance of evolutionary algorithms. In this implementation, this is performed by using a partition-distance problem approach (Gusfield, 2002). Measuring the similarity of mTSPs solutions as a partition-distance problem approach is an idea initially discussed and introduced by Cornejo-Acosta (2024).

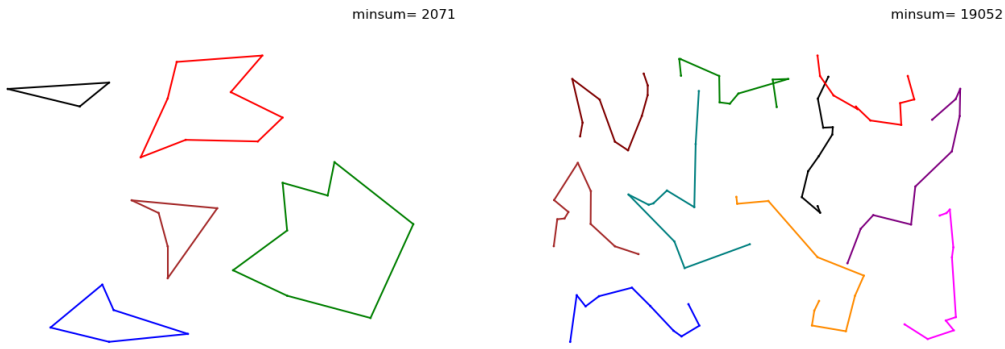
According to the nomenclature proposed by Cornejo-Acosta et al. (2023), and the features described above, this `AC-PGA` implementation can be used to address the following four

DFmTSP variants:

- Closed-path depot-free multiple traveling salesperson problem (CP-DFmTSP) with `minsum` objective function.
- Closed-path depot-free multiple traveling salesperson problem (CP-DFmTSP) with `minmax` objective function.
- Open-path depot-free multiple traveling salesperson problem (OP-DFmTSP) with `minsum` objective function.
- Open-path depot-free multiple traveling salesperson problem (OP-DFmTSP) with `minmax` objective function.

The execution of the algorithm can be configured through a plain-text configuration file that must be provided in advance. The software was designed such that many parameters can be set and the algorithm's behavior can be controlled; for example, stopping criteria, time limit, number of generations, objective function, type of path, number of salespersons, bounding constraints, input graph (instance), etc. The software supports only symmetric instances with all of the edge weight types of the TSPLIB formats (`EUC_2D`, `CEIL_2D`, `ATT`, `GEO`, and `EXPLICIT`). Once the algorithm is executed, an output file with the best-found solution and additional details is generated. For the updated list of input parameters and output details, please check the repository instructions and documentation (https://gitlab.com/alex.ca/AC-PGA).

The software was implemented in the `C++` programming language to prioritize efficiency, and uses the `cmake` tool to facilitate source code building. Also, the repository contains a `Python` script that can be used to visualize the solutions computed by the software. Figure 1 shows two examples of computed solutions. Note that only solutions of TSPLIB instances that contain display data (`TWOD_DISPLAY` or `COORD_DISPLAY`) or with edge weight type `EUC_2D` can be visualized. In order to handle the TSPLIB symmetric instances, `C++` and `Python` parsers are also provided as part of the software.



CP-DFmTSP with $m = 5$. OP-DFmTSP with $m = 10$.

Figure 1: Example of two solutions of the bays29 and kroA100 instances, respectively.

Finally, some improvements could be implemented in this software in the future. For instance, the software could be extended to approach other variants of mTSPs, specifically the variants that consider depots, such as the single-depot mTSP (SmTSP) or the multiple-depot mTSP (MmTSP). Another possible improvement is the parallelization of some sections of the algorithm, such as the ACO algorithm executions. This may result in significant performance improvement regarding running times.

Statement of need

Although numerous papers have been published on optimization problems related to the TSP and similar topics, many authors do not provide their software implementations. As a result, researchers working in the same area who wish to reproduce the computational experiments described in these papers must implement the algorithms from scratch. This process can be complex and very time-consuming, not only due to the inherent complexity of the methods proposed, but also because various technical and programming challenges may arise.

This software provides a flexible and efficient implementation of the `AC-PGA` metaheuristic, which was adapted to solve the main variants of depot-free multiple traveling salesperson problems. Thus, the general target audience includes researchers, students, and practitioners in the fields of operations research and computer science interested in working with optimization and routing problems related to the DFmTSPs and HpMP. In particular, the software is intended to be used by researchers working on the development of novel metaheuristics for routing problems, so they can benchmark their proposals against a state-of-the-art metaheuristic. Therefore, we expect the software to be useful for validating new proposals and algorithms, making it possible to advance and accelerate the state-of-the-art in routing optimization.

The software is designed to be user-friendly and easy to use, making it accessible to researchers with varying levels of expertise in optimization algorithms. The software is primarily intended for academic and research purposes; however, due to its open-source license, it can also be integrated into business applications to address real-life challenges in the fields of logistics, transportation, scheduling, and more.

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